

Calendar / Planner – PRD

1. Summary

Calendar / Planner module for managing schedules, events, to-dos, and reminders in a unified calendar view. Integrates with existing modules (Notes, Payments, Debts) so items from those modules appear on the calendar.

2. Goals

- Monthly/weekly/daily calendar view
- Create & manage one-time or recurring events
- To-do list with priority, deadline, and status
- Reminders via database & push notifications
- Integration: show Payments, Debts, Notes deadlines on calendar
- CRUD API for external frontend integration

3. User Stories

ID	Story
US-1	I can view a monthly calendar with events & tasks
US-2	I can create events with title, description, time, category
US-3	I can create recurring events (daily/weekly/monthly/yearly)
US-4	I can create to-dos with deadline, priority, and checklist
US-5	I can mark to-dos as complete
US-6	I get database notifications before events/tasks
US-7	I can see Payments & Debts deadlines on calendar
US-8	I can filter view by category or source module
US-9	I can drag & drop events to another date
US-10	I can export calendar to iCal format

4. Database Schema

4.1. calendar_events

Table for one-time and recurring events.

Column	Type	Constraints	Notes
id	bigIncrements	PK	
user_id	foreignId	constrained, cascade	
title	string(255)	required	
description	text	nullable	
location	string(255)	nullable	
start_at	datetime	required	
end_at	datetime	nullable	null = no duration set
is_all_day	boolean	default false	
category_id	foreignId	nullable, constrained -> calendar_categories	
color	string(9)	nullable	hex color #RRGGBB
recurrence_type	enum	nullable	daily, weekly, monthly, yearly
recurrence_interval	smallInteger	default 1, nullable	every X days/weeks/months/years
recurrence_end_at	date	nullable	when recurring stops
recurring_event_id	foreignId	nullable, self-referencing	parent event for recurring series
source_type	string(50)	nullable	polymorphic: payment, debt, note
source_id	bigInteger	nullable	polymorphic ID
metadata	json	nullable	extra data
created_at	timestamp		
updated_at	timestamp		
deleted_at	timestamp	nullable	soft deletes

Indexes:

- `user_id + start_at` for calendar queries
- `source_type + source_id` for polymorphic lookup

4.2. calendar_categories

Column	Type	Constraints	Notes
id	bigIncrements	PK	
user_id	foreignId	constrained, nullable	null = system default
name	string(100)	required	
color	string(9)	required	hex color #RRGGBB
is_default	boolean	default false	
created_at	timestamp		
updated_at	timestamp		

4.3. calendar_todos

Column	Type	Constraints	Notes
id	bigIncrements	PK	
user_id	foreignId	constrained	
event_id	foreignId	nullable, -> calendar_events	link to event
title	string(255)	required	
description	text	nullable	
priority	enum	required	low, medium, high
due_at	datetime	nullable	
completed_at	datetime	nullable	null = not completed
sort_order	smallInteger	default 0	
created_at	timestamp		
updated_at	timestamp		
deleted_at	timestamp	nullable	soft deletes

4.4. calendar_reminders

Column	Type	Constraints	Notes
id	bigIncrements	PK	
user_id	foreignId	constrained	
event_id	foreignId	nullable, -> calendar_events	
todo_id	foreignId	nullable, -> calendar_todos	

Column	Type	Constraints	Notes
remind_at	datetime	required	trigger time
reminded_at	datetime	nullable	sent time
created_at	timestamp		
updated_at	timestamp		

5. Models

Model	File	Notes
CalendarEvent	app/Models/CalendarEvent.php	<code>#[ObservedBy([CalendarEventObserver::class])]</code>
CalendarCategory	app/Models/CalendarCategory.php	<code>#[ObservedBy([CalendarCategoryObserver::class])]</code>
CalendarTodo	app/Models/CalendarTodo.php	<code>#[ObservedBy([CalendarTodoObserver::class])]</code>
CalendarReminder	app/Models/CalendarReminder.php	<code>#[ObservedBy([CalendarReminderObserver::class])]</code>

Relations:

- `CalendarEvent` belongsTo `User` , belongsTo `CalendarCategory` (nullable), morphTo `source`
- `CalendarEvent` hasMany `CalendarTodo`
- `CalendarEvent` hasMany `CalendarReminder`
- `CalendarCategory` belongsTo `User`
- `CalendarTodo` belongsTo `User` , belongsTo `CalendarEvent` (nullable)
- `CalendarReminder` belongsTo `User` , belongsTo `CalendarEvent` (nullable), belongsTo `CalendarTodo` (nullable)

6. Enums

Enum	File	Values
RecurrenceType	app/Enums/RecurrenceType.php	Daily, Weekly, Monthly, Yearly
TodoPriority	app/Enums/TodoPriority.php	Low, Medium, High
CalendarSource	app/Enums/CalendarSource.php	Payment, Debt, Note, Manual

7. Observers

Observer	File	Events
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Observer	File	Events
CalendarEventObserver	app/Observers/CalendarEventObserver.php	creating: generate code
CalendarCategoryObserver	app/Observers/CalendarCategoryObserver.php	creating: generate code
CalendarTodoObserver	app/Observers/CalendarTodoObserver.php	creating: generate code
CalendarReminderObserver	app/Observers/CalendarReminderObserver.php	creating, created: send notif

- `CalendarReminderObserver::created()` → send Filament database notification
- `CalendarEventObserver::creating()` → generate code via `getCode('calendar_event')`
- `CalendarTodoObserver::creating()` → generate code via `getCode('calendar_todo')`

8. Filament Resources

Resource	Navigation Group	Icon
CalendarEventsResource	Calendar	Heroicon::OutlinedCalendar
CalendarTodosResource	Calendar	Heroicon::OutlinedCheckCircle
CalendarCategoriesResource	Calendar	Heroicon::OutlinedTag

CalendarEventsResource - Layout

Form:

- Title, Description (RichEditor), Location
- Start at, End at, Is all day
- Category (select), Color (picker)
- Recurrence: type (select), interval, end at
- Source: read-only polymorphic link (if from another module)

Table:

. (row index)

-
- Code (copyable)
 - Title
 - Start at → end at
 - All day (icon)
 - Category (color badge)
 - Recurrence (badge)
 - Source (link to origin module)

- Created at
- Updated at

CalendarTodosResource - Layout

Form:

- Title, Description
- Related event (select, nullable)
- Priority (select enum)
- Due at
- Completed at (toggle, auto-set on check)

Table:

. (row index)

- Code (copyable)
- Title
- Priority (icon/color badge)
- Due at
- Status (completed icon)
- Event (link)
- Created at

CalendarCategoriesResource - Layout

Form:

- Name, Color (picker)
- Is default (toggle)

Table:

- Name
- Color (visual swatch)
- Is default (icon)
- Event count

9. Relation Managers

- `CalendarEventsResource` → `TodosRelationManager` (reuse `CalendarTodosTable`)
- `CalendarTodosResource` → `EventRelationManager`

10. API Routes

File: `routes/api/calendar.php` (require from `routes/api.php`)

GET	/api/events	# index (with ?month=, ?category=, ?source=)
POST	/api/events	# store
GET	/api/events/{event}	# show
PUT	/api/events/{code}	# update
DELETE	/api/events/{code}	# soft delete
DELETE	/api/events/{id}/force	# force delete
POST	/api/events/{id}/restore	# restore
PATCH	/api/events/{id}/duplicate	# duplicate event
GET	/api/todos	# index
POST	/api/todos	# store
GET	/api/todos/{todo}	# show
PUT	/api/todos/{code}	# update
DELETE	/api/todos/{code}	# soft delete
PATCH	/api/todos/{id}/toggle	# toggle completed
GET	/api/calendar/categories	# index
POST	/api/calendar/categories	# store
PUT	/api/calendar/categories/{id}	# update
DELETE	/api/calendar/categories/{id}	# delete
GET	/api/calendar/upcoming	# upcoming events+todos (next 7/14/30 days)
GET	/api/calendar/export	# export iCal

11. Controllers

Controller	File
Api\CalendarEventController	app/Http/Controllers/Api/CalendarEventController.php
Api\CalendarTodoController	app/Http/Controllers/Api/CalendarTodoController.php
Api\CalendarCategoryController	app/Http/Controllers/Api/CalendarCategoryController.php

12. Services (optional)

Service	File	Notes

Service	File	Notes
CalendarRecurrenceService	app/Services/CalendarRecurrenceService.php	Generate recurring instances
CalendarReminderService	app/Services/CalendarReminderService.php	Schedule & send reminders
CalendarIntegrationService	app/Services/CalendarIntegrationService.php	Pull data from Payments, Debts, Notes

13. Integration with Existing Modules

Payments

- When Payment created, auto-create `CalendarEvent` with `source_type=payment`.
- When Payment deleted, delete related event.

Debts

- When DebtInstallment created, auto-create `CalendarEvent` with `source_type=debt`.
- Installment deadlines appear on calendar.

Notes

- Optional: Note with `is_pinned=true` can become event via "Add to calendar" button.
- Notes with dates can appear as all-day events.

Implementation via existing module observers:

- `PaymentObserver::created()` → call `CalendarIntegrationService::sync(payment)`
- `DebtInstallmentObserver::created()` → call `CalendarIntegrationService::sync(installment)`

14. Recurring Logic

- `CalendarRecurrenceService::expand($event, $start, $end)` :
 1. Read `recurrence_type`, `interval`, `recurrence_end_at`
 2. Generate virtual instances between `$start` and `$end`
 3. Each instance has `original_start_at` (calculated from pattern)
 4. Instances are virtual (not stored in DB) - cached or calculated at runtime
 5. Override instances (change title/time for one occurrence) stored as `CalendarEvent` with `recurring_event_id` pointing to parent

15. Reminder / Notification

- `CalendarReminderService::schedule($event, $minutesBefore)` :
 1. Create `CalendarReminder` record with `remind_at = start_at - minutesBefore`
 2. `CalendarReminderObserver::created()` → `dispatch SendCalendarReminderJob`
 3. Job: send Filament database notification + push notification

Default reminder times:

- 15 minutes before
- 30 minutes before
- 1 hour before
- 1 day before (for all-day events)

16. Migration Order

1. `create_calendar_categories_table`
2. `create_calendar_events_table`
3. `create_calendar_todos_table`
4. `create_calendar_reminders_table`

17. Seeders

- `CalendarCategorySeeder` : default categories (Personal, Work, Health, Finance, Other)
- `CalendarEventSeeder` : sample events for development
- `CalendarTodoSeeder` : sample todos for development

18. Custom Filament View

Main Calendar page → use `CalendarWidget` (Livewire component) on dashboard or dedicated page:

- Monthly view (7-column grid)
- Prev/next month navigation
- Click date → see events & todos for that day
- Click event → edit modal
- Drag event → change date (API PATCH)

Implementation details TBD - custom page or widget.

19. Implementation Priority

Phase	Item
1	Models + migrations + enums + observers
2	Filament resources (CRUD events, categories, todos)
3	API controllers + routes
4	Recurring service + integration service (Payments, Debts, Notes)
5	Reminder service + notification
6	Calendar custom page / widget
7	Export iCal + drag & drop

20. Generate Alias Table

Alias	Model
calendar_event	CalendarEvent
calendar_todo	CalendarTodo

Add seeder for these aliases in `GenerateSeeder` or a separate seeder.

Constraints & Notes

- All models use SoftDeletes
- Use `#[ObservedBy]` attribute, do not register in provider
- 2-space indentation, aligned assignments
- API routes required from `routes/api.php`
- All file uploads via `Storage::disk('public')`
- Timezone: Asia/Jakarta
- Codes via `getCode('alias')`
- Polymorphic relation for cross-module integration